PATIENT-PORTAL BIOMARKER INGESTION

Smart acoustic audio and webcam facial distress tracking

Continuous edge pipelines convert spontaneous microphone audio and camera frames into quantified clinical biomarkers and real-time digital twin adaptation.

STREAM 04 • ACOUSTIC SENSING

Smart Acoustic Cough & Wheeze Sensor

Live microphone audio analyzed via Google YamNet (AudioSet Class 47/50) & Librosa DSP

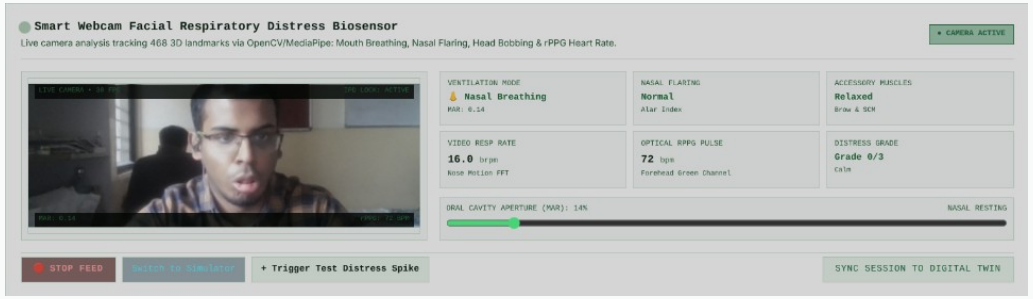


- 512-bin STFT Spectrum: Real-time frequency power & transient cough burst detection
- Tracked Biomarkers: Hourly Cough Count (0 bursts), Graded Severity (0–3), Wheeze (400–1600 Hz)
- Prospective Risk Ingestion: Real-time 48h flare-up risk calculation (62.4% High Risk outlook)

STREAM 05 • FACIAL COMPUTER VISION

Smart Webcam Facial Distress Biosensor

Live webcam analysis tracking 468 3D landmarks via OpenCV/MediaPipe + optical rPPG



- 468 3D Landmark Mesh: Real-time 30 FPS facial geometry with IPD stabilization lock
- Tracked Biomarkers: Oral Breathing (MAR: 0.14 vs >0.35), Alar Flaring, SCM Accessory Strain
- Contactless Vitals: Forehead green-channel pulse (72 bpm) & nose-motion resp rate (16.0 brpm)

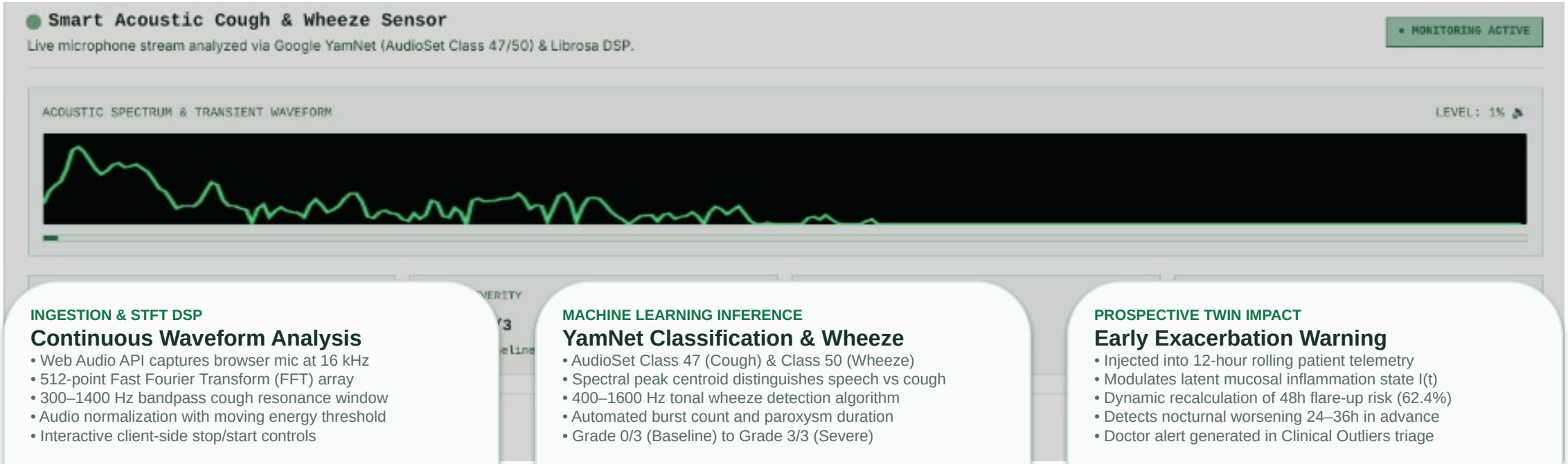
CLOSED-LOOP DIGITAL TWIN INTEGRATION

Acoustic cough bursts and facial respiratory distress telemetry stream into the 12-hour rolling wearable feature window, modulate the latent mucosal inflammation state I(t) in the Euler-Maruyama ODE, and trigger clinician outlier retraining via Elastic Weight Consolidation (EWC).

BIOACOUSTIC SIGNAL PROCESSING

Acoustic frequency spectrum and transient cough burst detection

Client-side Web Audio API and Google YamNet DSP extract paroxysmal cough frequency and wheeze signatures to detect early mucosal narrowing without hospital spirometry.



INGESTION & STFT DSP

Continuous Waveform Analysis

- Web Audio API captures browser mic at 16 kHz
- 512-point Fast Fourier Transform (FFT) array
- 300–1400 Hz bandpass cough resonance window
- Audio normalization with moving energy threshold
- Interactive client-side stop/start controls

MACHINE LEARNING INFERENCE

YamNet Classification & Wheeze

- AudioSet Class 47 (Cough) & Class 50 (Wheeze)
- Spectral peak centroid distinguishes speech vs cough
- 400–1600 Hz tonal wheeze detection algorithm
- Automated burst count and paroxysm duration
- Grade 0/3 (Baseline) to Grade 3/3 (Severe)

PROSPECTIVE TWIN IMPACT

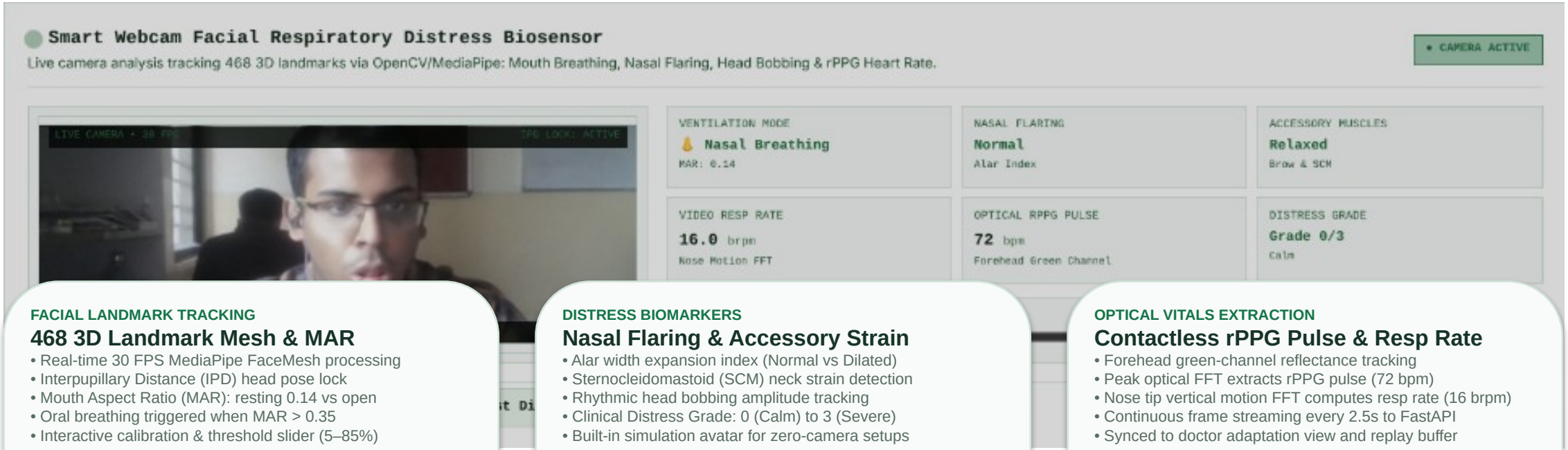
Early Exacerbation Warning

- Injected into 12-hour rolling patient telemetry
- Modulates latent mucosal inflammation state I(t)
- Dynamic recalculation of 48h flare-up risk (62.4%)
- Detects nocturnal worsening 24–36h in advance
- Doctor alert generated in Clinical Outliers triage

FACIAL BIOMETRICS & OPTICAL DSP

Optical facial geometry, mouth breathing, and rPPG vital signs

MediaPipe 468 3D landmark mesh and optical green-channel rPPG extract breathing pattern changes, accessory muscle recruitment, and pulse from ordinary webcam video.



FACIAL LANDMARK TRACKING

468 3D Landmark Mesh & MAR

- Real-time 30 FPS MediaPipe FaceMesh processing
- Interpupillary Distance (IPD) head pose lock
- Mouth Aspect Ratio (MAR): resting 0.14 vs open
- Oral breathing triggered when MAR > 0.35
- Interactive calibration & threshold slider (5–85%)

DISTRESS BIOMARKERS

Nasal Flaring & Accessory Strain

- Alar width expansion index (Normal vs Dilated)
- Sternocleidomastoid (SCM) neck strain detection
- Rhythmic head bobbing amplitude tracking
- Clinical Distress Grade: 0 (Calm) to 3 (Severe)
- Built-in simulation avatar for zero-camera setups

OPTICAL VITALS EXTRACTION

Contactless rPPG Pulse & Resp Rate

- Forehead green-channel reflectance tracking
- Peak optical FFT extracts rPPG pulse (72 bpm)
- Nose tip vertical motion FFT computes resp rate (16 brpm)
- Continuous frame streaming every 2.5s to FastAPI
- Synced to doctor adaptation view and replay buffer